

Logistics Warehouse EDA Report

NexAfrica Internship Task

3204 Items Analysis

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SQL File 3* SQL File 3* x

Limit to 1000 rows

```
1 • SELECT AVG(stock_level) AS avg_stock,  
2     MIN(stock_level) AS min_stock,  
3     MAX(stock_level) AS max_stock,  
4     AVG(daily_demand) AS avg_demand  
5 FROM logistics_dataset;
```

Result Grid

Filter Rows: Export: Wrap Cell Content:

	avg_stock	min_stock	max_stock	avg_demand
▶	263.4916	20	499	25.43586766541822

The Warehouse holds average 263 units per item, demand is 25 per day.

SQL File 3*SQL File 3*SQL File 4*

</

Pharma holds the most stock – needs temperature control and careful handling

SQL File 3*SQL File 3*SQL File 4*SQL File 5* x

Automotive highest demand: 25.99 per day

Electronics lowest: 24.71

So demand is balanced across all categories – good for warehouse planning.

SQL File 3*			
SQL File 3*			
SQL File 4*			
SQL File 5*			
SQL File 6* x			
Limit to 1000 rows			
<pre> 1 • SELECT item_id, category, item_popularity_score 2 FROM logistics_dataset 3 ORDER BY item_popularity_score 4 LIMIT 10;</pre>			
Result Grid			
Filter Rows:			
Export:			
Wrap Cell Content:			
Fetch rows:			
item_id	category	item_popularity_score	
ITM12854	Automotive	0.1	
ITM12041	Groceries	0.1	
ITM12254	Apparel	0.1	
ITM10430	Apparel	0.1	
ITM12980	Groceries	0.1	
ITM11313	Electronics	0.1	
ITM10804	Apparel	0.1	
ITM12003	Electronics	0.1	
ITM12008	Apparel	0.1	
ITM12242	Pharma	0.1	

Top 10 least popular items all score 0.1

Mix of Apparel, Groceries, Automotive. These are slow-moving stock, consider discounting or relocating to less accessible zones.

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

1 • SELECT item_id, category, stock_level, reorder_point
2 FROM logistics_dataset
3 WHERE stock_level <= reorder_point
4 ORDER BY stock_level
5 LIMIT 15;

```

Result Grid Filter Rows: Export: Wrap Cell Content: Fetch rows:

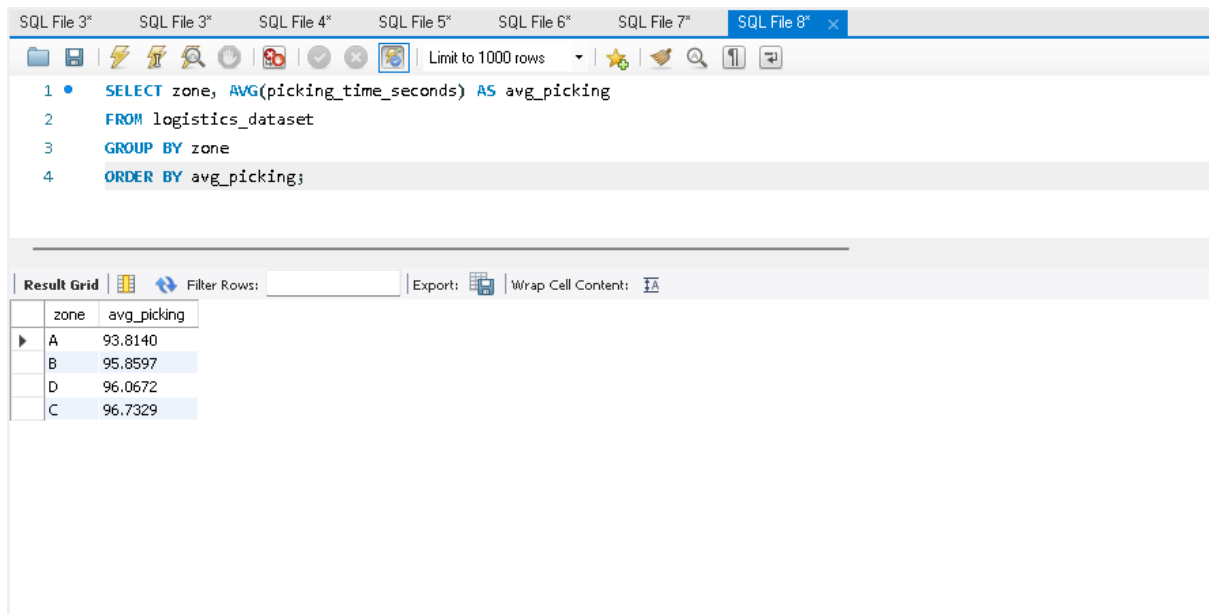
	item_id	category	stock_level	reorder_point
▶	ITM12143	Electronics	20	38
	ITM10011	Groceries	20	66
	ITM10804	Apparel	20	62
	ITM11554	Electronics	21	30
	ITM10886	Apparel	21	40
	ITM12375	Automotive	21	21
	ITM10988	Groceries	21	35
	ITM12622	Pharma	22	99
	ITM10723	Pharma	22	88
	ITM10267	Electronics	22	76
	ITM11458	Electronics	22	53
	ITM10471	Automotive	22	61
	ITM10822	Apparel	22	88
	ITM11203	Electronics	22	79
	ITM11685	Apparel	22	57

logistics_dataset 1 x

Output

15 items are below reorder point, with lowest stock at 20 units – immediate purchase orders required.

Zone A is most efficient – recommend storing high-demand items in Zone A to reduce picking time.



The screenshot shows a SQL IDE interface with multiple tabs labeled 'SQL File 3*' through 'SQL File 8*'. The active tab is 'SQL File 8*'. The query editor contains the following SQL code:

```
1 • SELECT zone, AVG(picking_time_seconds) AS avg_picking
2 FROM logistics_dataset
3 GROUP BY zone
4 ORDER BY avg_picking;
```

Below the query editor, there is a toolbar with icons for file operations, a 'Limit to 1000 rows' dropdown, and a star icon. The 'Result Grid' tab is active, displaying the following data:

	zone	avg_picking
▶	A	93.8140
	B	95.8597
	D	96.0672
	C	96.7329

At the bottom of the result grid, there is a toolbar with icons for 'Filter Rows', 'Export', and 'Wrap Cell Content'.

Top 10 items cost R5 per unit to handle – review packaging to reduce cost

The screenshot shows a database management interface with a Navigator on the left, a SQL editor in the center, and a Result Grid at the bottom.

Navigator: Shows a tree view of the database structure. The 'warehouse_db' schema is selected, showing tables, views, stored procedures, and functions. The 'world' schema is also visible.

SQL Editor: Contains the following SQL query:

```
1 • SELECT item_id, category, handling_cost_per_unit
2 FROM logistics_dataset
3 ORDER BY handling_cost_per_unit DESC
4 LIMIT 10;
5
6
7
```

Result Grid: Displays the top 10 items based on handling cost per unit. The columns are 'item_id', 'category', and 'handling_cost_per_unit'.

item_id	category	handling_cost_per_unit
ITM11284	Automotive	5
ITM10148	Apparel	5
ITM11980	Pharma	5
ITM12772	Groceries	5
ITM11797	Automotive	5
ITM12991	Automotive	4.99
ITM11968	Apparel	4.99
ITM10143	Pharma	4.99
ITM11344	Apparel	4.99
ITM10729	Electronics	4.99

Administration: Shows the 'Schemas' tab selected.

Information: Shows the 'Schema: warehouse_db' selected.

Apparel reorders fastest (8.41 days),

Electronics slowest (8.71 days)

All categories 8-9 days cycle.

The screenshot shows a database management interface. On the left, a 'SCHEMAS' panel lists databases: sakila, sys, warehouse_db (selected), and world. The 'warehouse_db' database is expanded, showing Tables, Views, Stored Procedures, and Functions. The main area displays a SQL query in a text editor:

```
1 • SELECT category, AVG(reorder_frequency_days) AS avg_freq
2 FROM logistics_dataset
3 GROUP BY category
4 ORDER BY avg_freq;
```

Below the query editor, a 'Result Grid' shows the results of the query. The grid has two columns: 'category' and 'avg_freq'. The results are sorted by 'avg_freq' in ascending order.

category	avg_freq
Apparel	8.4100
Automotive	8.4255
Pharma	8.4288
Groceries	8.5599
Electronics	8.7143

At the bottom of the interface, there are tabs for 'Administration' and 'Schemas', and a status bar indicating 'Schema: warehouse_db'.